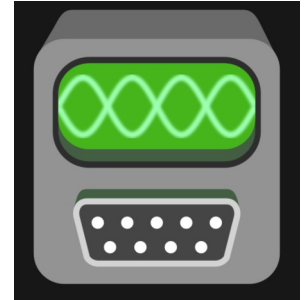


BNO055 Procedure

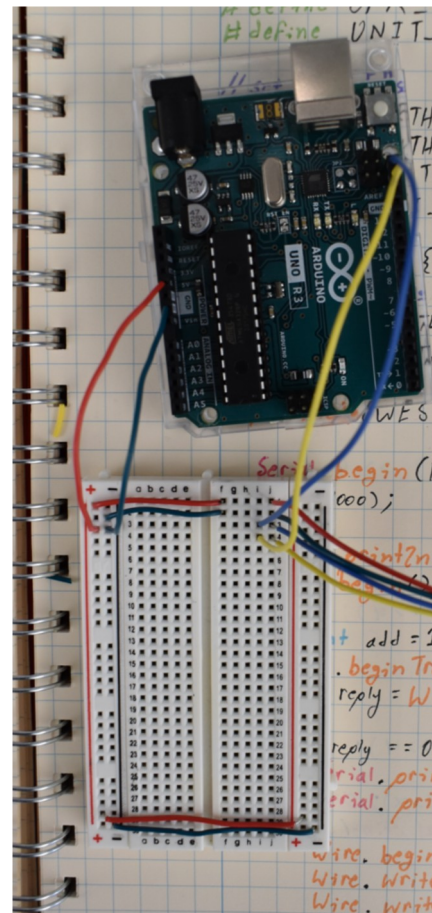
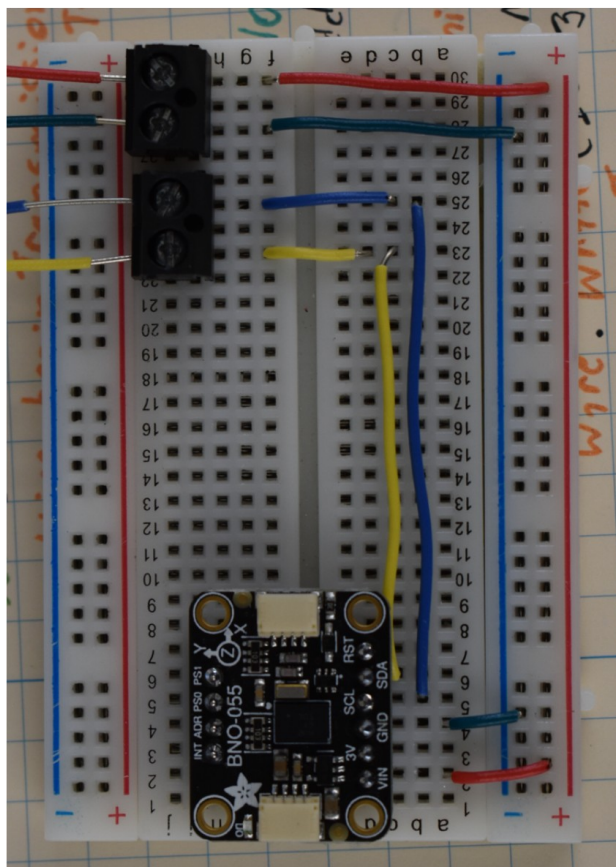
1. SerialPlot

Download the SerialPlot software (<https://hackaday.io/project/5334-serialplot-realtime-plotting-software>). It should be the first or second link when you look up “SerialPlot” with your search engine of choice. This step is not strictly necessary, but it’s helpful for visualizing sensor output.



2. Circuit Part 1

The connections on this circuit are wonderfully simple. 5V goes to Vin, ground to ground, SDA to SDA, and SCL to SCL¹. We’re working with a big-kid sensor now, so your wires should NOT look like spaghetti-junction². You’ll set up your BNO055 on one breadboard, connect those wires to a second breadboard, and connect the wires on the second breadboard to your Arduino. In a later part of the lab, we’ll be connecting a circuit to the second breadboard that we’ll want to keep stationary while still being able to move the BNO055.



- 1 Does the straightforward labeling not bring a tear to your eye??? It's beautiful. :')
- 2 You may have been told that the most important law in electronics is $V = IR$. In reality, it's $r = p * c$. Meaning circuit-reliability = prettiness*conscientiousness. The prettier your circuit, the better it will work!!

I use terminal blocks here. You don't have to do this, but I think it's a nice touch. The main down-side of terminal blocks is that they WILL ABSOLUTELY fall out of your breadboard if they aren't soldered. Adding some zip-ties is an easy solution to this.

Before we move on to some programming, let's check our connections with the DMM. When you have this many connections, a quick check after wiring everything up can save *a lot* of headache later³.

3. I2C scanner

Typically, you'll be able to find the register for your sensor in the data-sheet, but not all documentation is made equal and not all sensors behave how you would expect. For many of you, the function we are writing is going to be a true life-long friend... Today, we are making an I2C scanner⁴. Not only is this function useful for finding chip registers, it's also helpful for seeing if any of your peripheral devices are actually responding!

The I2C scanner loops through 127 possible addresses and attempts to contact a peripheral at each address. When it reaches a peripheral device returns an acknowledgment, the I2C scanner returns the appropriate address.

4. Getting Data

Before we actually get any data from this dude, we gotta change its settings (exciting, I know). I encourage you to look through the BNO055 data-sheet to try and find these settings yourself as an exercise, but for the purpose of this lesson, you have a handout with the relevant registers.

Step 1: Change the operating mode

Write xxxx1000b to 0x3D

Step 2: Change units to degrees

Write xxxxx0xxb to 0x3B.

Step 3: Write Data Register

Euler angle X is at register 0x1A. Each angle is 2 bytes, and angle Y starts at 0x1C, and angle Z starts at 0x1E. This means to get every angle, we can simply write 0x1A then read in 6 bytes.

Step 4: Read

For each angle read (1) the LSB, and (2) the MSB. We must shift the MSB 8 bits to the left, then combine with the LSB using a bitwise OR. Then, we divide by 16 to get degrees. Shifting to the right by 4 is equivalent to dividing by 2 to the power of 4 (each right shift halves the binary number), or 16.

So, to get our number in degrees we do $((\text{MSB} \ll 8) | \text{LSB}) \gg 4$

3 You should be thinking about other quick "check-ups" you can do on your hardware and software as you work on larger projects. A few seconds of checking your work can save you hours of future troubleshooting (and several trips to the wasted hours counter).

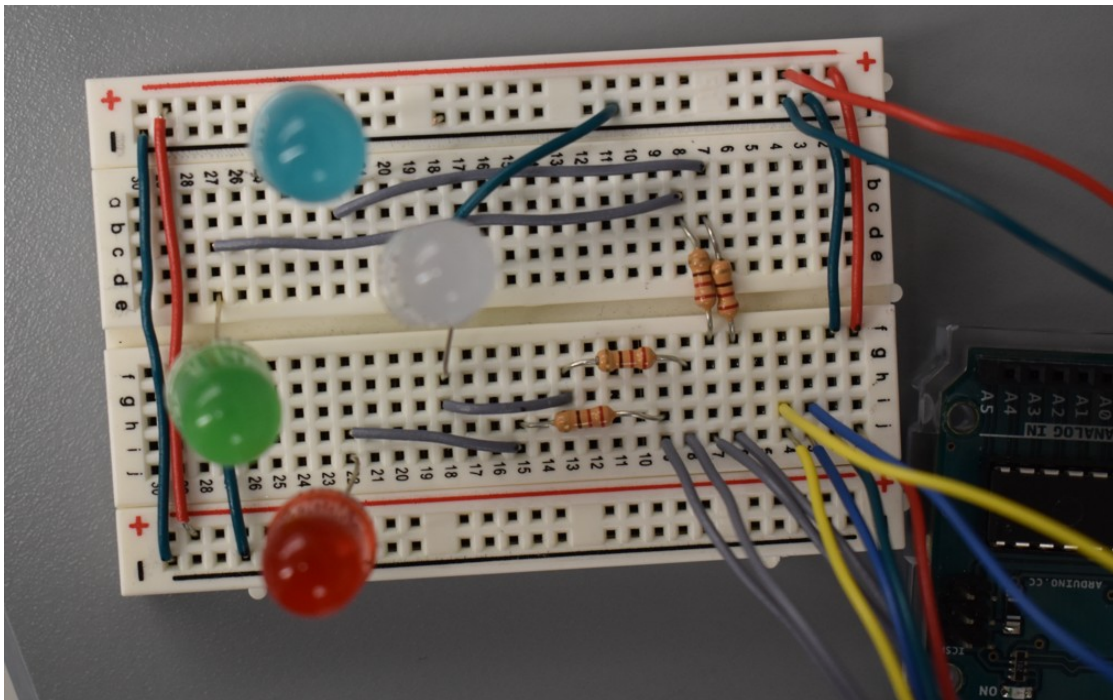
4 I'm sure we had to pause here due to all the shouts of excitement and applause from the crowd.

Step 5: Visualizing the Data

Make sure your serial monitor is closed. Open the SerialPlot app and adjust the parameters so you can graph all 3 angles.

5. Circuit Part 2

Now that we can get heading data, we're going to practice using it. With the kind of work we do, what's the best way to test something new?? A blinky light of course!!! Let's set up four LEDs corresponding to four directions. Then, we'll write some if-else statements to light up LEDs based on the tilt of the BNO.



Post Lesson Challenges:

1. Make an I2C scanner that can detect multiple peripheral devices.
2. Make functions for the LED control code.
3. The BNO055 offers more than orientation data. Try to access a different data register.